

Chun (Gene) Yu

28 Harston Walk, London | gene.yu.tw@gmail.com | 07904 548 316 | linkedin.com/in/gene-yu-tw | Portfolio

Education

Imperial College London, UK

Sep. 2023 – Sep. 2024

MSc in Computing, with Distinction

- Coursework: Software Systems Engineering, Machine Learning, Computational Finance, Distributed Systems
- Master Thesis: “Robust Time Series Causal Discovery for Agent-Based Model Validation”

National Tsing Hua University, Taiwan

Sep. 2018 – Jun. 2022

BS in Interdisciplinary Program of Engineering, (Top 4%, Dean's List 2x)

- Specialised in Electrical Engineering and Power Mechanical Engineering with a focus on Electrical Control
- Awards: 2nd Place, College of EECS Project Competition

Experience

Software Engineer – Trading Systems, xSyphon, London, UK

Apr. 2025 – Present

- Built a natural-language query interface with OpenAI API, enabling SQL-based trading data insights via context-aware chat sessions
- Developed a markup coordinator aggregating liquidity provider quotes, dealer markups, and positions, computing AI-driven markups every minute and pushing snapshots to the trading gateway
- Maintained CI/CD pipelines and multi-environment releases, leveraging AWS CloudWatch for monitoring and managing infrastructure configurations

Summer Research Intern, Academia Sinica – Taipei, Taiwan

Jul. 2022 – Aug. 2022

Natural Language Processing and Sentiment Analysis Lab

- Collaborated with a research team on an AI-Automated Training System for Figure Skating, concentrating on improving temporal video alignment by incorporating posture features

Projects

Robust Consensus-Driven Framework for Time Series Causal Discovery

May 2024 – Sep. 2024

- Proposed Robust Consensus-Driven framework that ensembles multiple causal discovery methods with robustness checks, achieving stable and accurate structure learning on noisy, high-dimensional time-series data
- Validated the framework across synthetic and fMRI benchmarks, demonstrating significant improvements in accuracy compared with single-method baselines (e.g., VarLiNGAM, PCMCI)

FoodHub Web Application (Software System Engineering)

Feb. 2023 – Mar. 2023

- Architected microservices using Azure and Supabase for scalability and database solutions
- Streamlined CI/CD pipelines to automate deployment processes across environments
- Developed Food Finder feature with secure user onboarding and recipe discovery APIs

IMU-Based Motion Analysis System - NTHU Reliable Computing Lab

Jan. 2021 – Jan. 2022

- Developed a motion-analysis system using 6-axis IMU sensors to reconstruct football kick trajectories.
- Achieved low RMS error in velocity and position; Authored the peer-reviewed publication: **Yu, C.**; Huang, T. Y.; Ma, H. P. Motion Analysis of Football Kick Based on an IMU Sensor. *Sensors* 2022, 22, 6244.

Technical Skills

Programming & Tools: Python, C++ , Java, JavaScript, HTML/CSS, React, Git, Verilog HDL, MATLAB

Cloud & DevOps: AWS (EC2, S3, Lambda), Azure, CI/CD, Docker

Certificates: AWS DevOps Specialization, Meta Front-End Developer (Coursera)

Extracurricular Activities

Exchange Student, Tel Aviv University – Israel

Feb. 2022 – Jun. 2022

- Conceptualised a pedestrian safety route app at 2022 Ignites International Hackathon

Tanzania Volunteer Team, Curriculum Planning Section – Taiwan

Nov. 2019 – Nov. 2020

- Organized software teaching lessons for kids and raised 120+ second-hand computers from businesses.